Eshita Banerjee

Nationality: Indian (She/ Her)

I am currently working as a Postdoctoral Fellow at Carnegie Observatories, Pasadena, California. My research focuses on the formation and evolution of high-redshift galaxies, particularly through quasar absorption-line analysis. I have primarily studied the extended gaseous environments surrounding galaxies at cosmic noon, as traced by both emission and absorption diagnostics.

Education

Postdoctoral Fellow — Pasadena, CA, US
Carnegie Observatories — Aug., 2026 – Present

- *Advisor:* Dr. Gwen Rudie (LARC Collaboration)

Ph.D. — IUCAA, Pune, India
Astronomy & Astrophysics — Oct., 2020 – July, 2026

- *Dissertation title*: Gaseous atmospheres of high redshift Lyα emitters
- *Advisor:* Dr. Sowgat Muzahid (Thesis submitted on Sep, 2025)

Master of Science — IIT Kharagpur, India
Physics — 2018 – 2020

Bachelor of Science — University of Calcutta, India
Physics — 2014 – 2017

Contact

Carnegie Observatories,
813 Santa Barbara St,
Pasadena, CA 91101

(+1) (626) 577-1122
eshitaban18@gmail.com
ebanerjee@carnegiescience.edu
Website
Linkedin
Github
0009-0002-7382-3078

Observational proposals

KMOS (VLT) (Co-I) — Connecting the ISM and CGM properties of galaxies at cosmic noon
Programme ID: 111.24ZE (PI- Dr. Sowgat Muzahid)

HST (Co-I) — Baryonic Escape, Abundances, and Thermodynamics in Cosmic Ecosystems
Cycle 33, GO time (PI- Prof. Sean Johnson)

Awards

2020 - 2026	**Junior & Senior research fellow**, IUCAA, Pune
2018-2020	**Fellowship for the integrated M.Sc.-Ph.D** program in department of physics Indian Institute of Technology (IIT), Kharagpur
2018-2020	Cleared national level exam **IIT-JAM**, **GATE**, **INAT**
2014-2017	**DST-INSPIER Scholarship**: For being in top 1 percentile in +2 examination

skills

Observational Data analysis

Optical	Multi Unit Spectroscopic Explorer (**MUSE**) and Ultraviolet and Visual Echelle-Spectrograph (**UVES**) on Very Large Telescope (**VLT**)
	High Resolution Echelle Spectrometer (**HIRES**) on Keck telescope
	HETDEX, Sloan Digital Sky Survey (**SDSS**)
Infrared	**Xshooter, KMOS**

Coding and Operating systems
Python, Latex, Linux, Git (advanced)
C, Fortran, Shell script (intermediate)
ADQL/SQL, HTML (basic)

Astronomy related software
Voigt profile fitting program **VPFIT**; Photoionization simulation code **CLOUDY**;
3D Spectroscopy **CubEx, QFitsView, Mpdaf, Pyplatefit**
Photometry **DS9, SEP**; Stellar population synthesis **BAGPIPES**, **pPXF**

Misc. Data analysis, High-performance computing, Science outreach and teaching

Language Proficient in English, Hindi and Bengali (native).

Conferences, Workshops and Schools

Invited Talks

Milan, Italy	COSMIB research group, University of Milano-Bicocca, Italy; Oct, 2024
CARL, France	Centre de Recherche Astrophysique de Lyon, Lyon observatory, France; Oct, 2024
AIP, Germany	Leibniz-Institute for Astrophysics Potsdam (AIP), Germany; Nov, 2024

Conference Talks

March, 2025	MUSE Busy-week conference (Virtual)
IUCAA, India	Conference: Baryons Beyond Galactic Boundaries (BBGB), Dec, 2024
CUHP, India	Conference on Advancements in AGN, Galaxy Cluster, and IGM Research, 2024
2021 & 2022	MUSE Busy-week conference (Virtual)

Flash talks & poster presentations

STScI, Baltimore	Spring symposium, 2024
IUCAA, India	Galflows, 2023
ARIES, India	Young Astronomers Meet, 2022
2022	PoSter-2022 (Virtual)

Workshops and Schools

IUCAA, India	Workshop on Machine learning in Astronomy and Astrophysics, 2023

IIA, India	Summar school on Astronomy & Astrophysics, Kodaikanal Solar Observatory, 2019

Professional activities and Outreach

Mentoring students	Adarsh Mahor (Summer project, 2022) Subhadeep Bhattarcharyay (M.Sc. dissertation, 2023-24)
September, 2025	**Teaching assistant** Course: Astronomical observational techniques Graduate school, IUCAA
January, 2025	**Scientific organizing committee** At a national level conference Young Astronomers' Meet (YAM) hosted by TIFR, India
May, 2024	**IUCAA summer school and refreshers course**
Topic	3D spectroscopy and hands-on sessions on MUSE data
February, 2023	**Local organizing committee** At a national level conference GALFLOW hosted by IUCAA, India
November, 2022	**School visit, Young Astronomers' Meet** Scientific talks and demonstrations to 500+ school students
2021-2025	**National Science Day** Actively participated in scientific talks, poster presentations and hands-on demonstrations to 20,000+ people at the National Science Day event at IUCAA, Pune, India

References

1. **Dr. Sowgat Muzahid** (sowgat@iucaa.in)
   IUCAA, Pune, Maharashtra, , India, PIN-411007

2. **Prof. Joop Schaye** (schaye@strw.leidenuniv.nl)
   Leiden Observatory, NL-2300 AA Leiden, the Netherlands

3. **Dr. Sean D Johnson** (seanjoh@umich.edu)
   University of Michigan, US

Publications of Eshita Banerjee (link to ADS Library)

1. **Eshita Banerjee**, Sowgat Muzahid, Joop Schaye, Sean D. Johnson and Sebastiano Cantalupo
MUSEQuBES: Investigating the Physical and Chemical Properties of Circumgalactic Gas Around Lyα Emitters at $z \approx 3.3$ [ArXiv] (*under journal review*)

2. **Eshita Banerjee**, Sowgat Muzahid, Joop Schaye, Jérémy Blaizot, Sean D. Johnson, Jorryt Matthee, Anne Verhamme, Nicolas Bouché, and Sebastiano Cantalupo
"MUSEQuBES: Connecting HI absorption with Ly-α emitters at $z \approx 3.3$"
ApJ, 2025, Vol. 980, Issue 2, id.171, 13 pp [ArXiv]

3. **Eshita Banerjee**, Sowgat Muzahid, Joop Schaye, Sebastiano Cantalupo and Sean D. Johnson
"MUSEQuBES: Unveiling Cosmic Web Filaments at $z \approx 3.6$ through Dual Absorption and Emission Line Analysis"
ApJ Letters, 2025, Vol. 979, Issue 2, id.L32, 6 pp. [ArXiv] [Press article link]

4. **Eshita Banerjee**, Sowgat Muzahid, Joop Schaye, Sean D. Johnson and Sebastiano Cantalupo
"MUSEQuBES: The relation between Lyα emitters and CIV absorbers at $z \approx 3.3$"
MNRAS, 2023, Vol. 524, Issue 4, pp.5148-5165 [ArXiv]